THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE TWO-CHAPTER TEST SET A

Chapters 9 & 10: **The Amazing World of Solutes, Solvents and Solutions, Light: Mirrors and Lenses** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A substance has mass 156 g and volume 20 cm³. Its density is: 1

(a) 0.128 g/cm³ **(b)** 7.8 g/cm³ **(c)** 3.9 g/cm³ **(d)** 15.6 g/cm³

2. In a salt–water solution, the solvent is: 1

(a) glass **(b)** water **(c)** salt **(d)** air

3. Solubility is the maximum amount of solute that dissolves in: 1

(a) any quantity at any temperature
(b) 1 kg of solute
(c) air only
(d) a fixed quantity (100 mL) of solvent at a given temperature

4. Fish need dissolved oxygen, which is less in: 1

(a) clean flowing water **(b)** ice
(c) warm water **(d)** cold water

5. Volume of a liquid is measured with a: 1

(a) clock **(b)** spring balance
(c) measuring cylinder **(d)** balance

6. A ray of light makes an angle of 65° with the normal to a plane mirror. The angle of reflection is: 1

(a) 130° **(b)** 65° **(c)** 25° **(d)** 115°

7. The angle of incidence is the angle between the incident ray and the: 1

(a) horizon **(b)** mirror surface **(c)** reflected ray **(d)** normal

8. The laws of reflection are valid for: 1

(a) only convex mirrors **(b)** only concave mirrors
(c) plane, concave and convex mirrors **(d)** only plane mirrors

9. A spherical mirror whose reflecting surface bulges outward is: 1

(a) concave lens **(b)** plane mirror **(c)** convex mirror **(d)** concave mirror

10. Which mirror is used as a rear-view mirror in vehicles? 1

(a) lens **(b)** concave mirror **(c)** convex mirror **(d)** plane mirror

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 10–15 words each.

11. Why does ice float on water? 2

12. Why is floating ice important for aquatic life? 2

13. Differentiate a convex lens and a concave lens. 2

14. Name the lens used in a magnifying glass. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Differentiate between solute–solvent in a solid–liquid and liquid–liquid solution. 3

16. Why do we see lateral inversion in all three types of mirrors? Give an example. 3

17. Why do fishes die in warm polluted water? 3

OR

What is the difference between mass and volume? Give units. 3

18. A plane mirror is 2 m away from you. How far is your image from you? 3

OR

What is a diverging mirror and a diverging lens? Give one example each. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. Describe the use of the measuring cylinder and balance with precautions to read the meniscus. 5

OR

Write about solid–liquid, liquid–liquid, gas–liquid and solid–solid solutions with examples. 5

20. Explain refraction with examples and how it makes a pencil in water appear bent. 5

OR

Explain how lenses are used in spectacles, magnifying glasses and cameras. 5

END OF QUESTION PAPER — Total: 40 Marks

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